Symmetry

SEE ALSO
❮ 86–87 Straight lines
Rotations 100–101 ❯
Reflections 102–103 ❯

THERE ARE TWO TYPES OF SYMMETRY—REFLECTIVE AND ROTATIONAL.

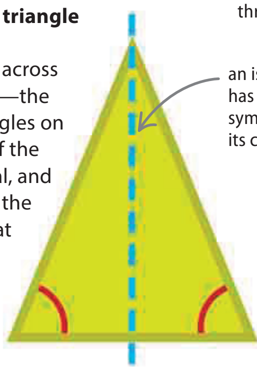
A shape has symmetry when a line can be drawn that splits the shape exactly into two, or when it can fit into its outline in more than one way.

Reflective symmetry

A flat (two-dimensional) shape has reflective symmetry when each half of the shape on either side of a bisecting line (mirror line) is the mirror image of the other half. This mirror line is called a line of symmetry.

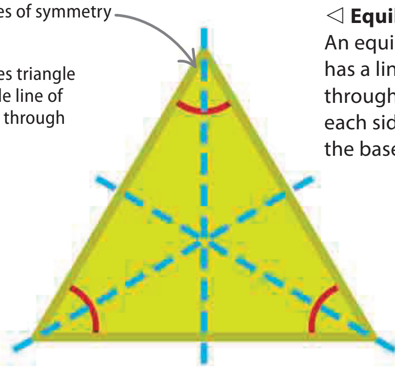
▷ **Isosceles triangle**
This shape is symmetrical across a center line—the sides and angles on either side of the line are equal, and the line cuts the base in half at right angles.

an isosceles triangle has a single line of symmetry through its center



Isosceles triangle

equilateral triangles have three lines of symmetry

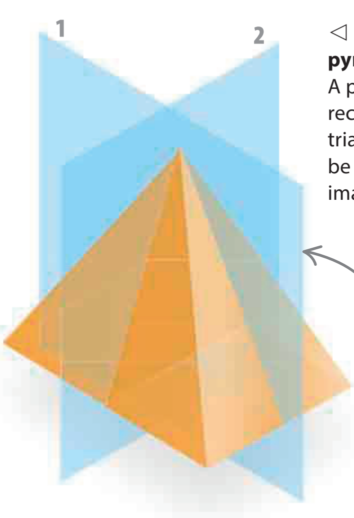


Equilateral triangle

◁ **Equilateral triangle**
An equilateral triangle has a line of symmetry through the middle of each side—not just the base.

Planes of symmetry

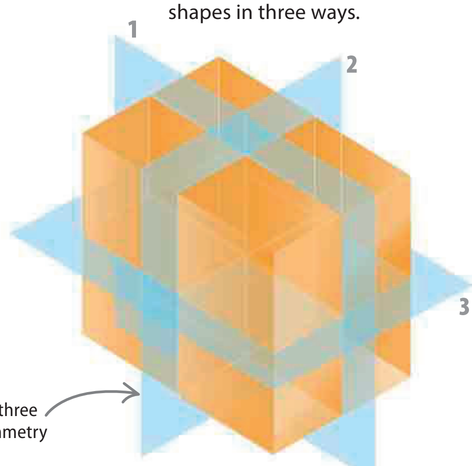
Solid (three-dimensional) shapes can be divided using "walls" known as planes. Solid shapes have reflective symmetry when the two sides of the shape split by a plane are mirror images.



◁ **Rectangle-based pyramid**
A pyramid with a rectangular base and triangles as sides can be divided into mirror images in two ways.

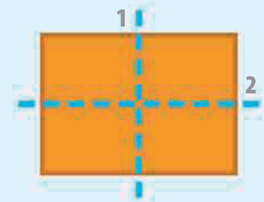
a rectangle-based pyramid has two planes of symmetry

▽ **Cuboid**
Formed by three pairs of rectangles, a cuboid can be divided into two symmetrical shapes in three ways.

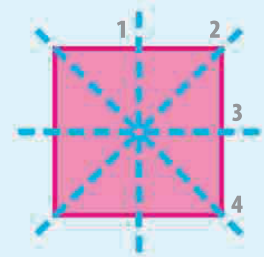


a cuboid has three planes of symmetry

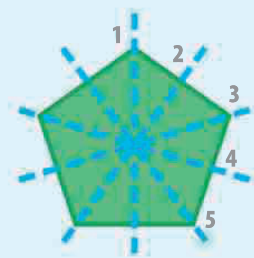
▽ **Lines of symmetry**
These are the lines of symmetry for some flat or two-dimensional shapes. Circles have an unlimited number of lines of symmetry.



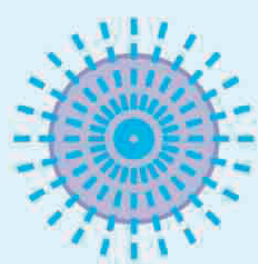
Lines of symmetry of a rectangle



Lines of symmetry of a square



Lines of symmetry of a regular pentagon



Every line through the middle of a circle is a line of symmetry